QUEUENOW — Ultimate Master Guide

Complete Reference Document Featuring `main`, `resources`, and Every Project File

1. Root Configuration & Build Files

File / Folder	Role & Responsibility
`pom.xml`	Defines Maven coordinates, dependencies (Spring Web, ZXing QR), and Java 17 target.
`mvnw` & `mvnw.cmd`	Maven Wrapper scripts enabling builds without global Maven installation.
`.gitignore` & `.gitattributes`	Excludes build artifacts (`target/`) and normalizes cross-platform line endings.
`HELP.md` & `queuenow.zip`	Spring documentation reference and initial template archive.

2. Hidden Wrapper Folder (`.mvn/`)

File / Folder	Role & Responsibility
`.mvn/wrapper/`	Directory containing Maven wrapper engine binaries and property settings.
`maven-wrapper.properties`	Locks down the exact Apache Maven download URL for uniform team builds.

3. The `src/main/` Master Directory

File / Folder	Role & Responsibility
`src/main/`	**Main Root Hub:** The core master folder containing all production application code, source packages, web templates, and configuration assets.
`src/main/java/`	**Java Source Container:** Holds all object-oriented backend packages and controllers.
`src/main/resources/`	**Resources Container:** Holds static frontend assets and application configuration properties.

4. Java Packages Inside `src/main/java/`

File / Folder	Role & Responsibility
`com/queuenow/`	Root namespace package where all team modules (controllers, services, models) are organized.
`QueuenowApplication.java`	Bootstrap class with `main()` method and `@SpringBootApplication` starting embedded Tomcat.
`target/` (Generated)	Output directory created during compilation holding packaged executable JAR files.

5. Deep Dive into `src/main/resources/` Subfolders

File / Folder	Role & Responsibility
`src/main/resources/`	Master resources hub bundled automatically into the final compiled Spring Boot JAR archive.
`static/`	**Static Subfolder:** Serves unedited public assets directly to browsers (CSS files, client JavaScript, pre-generated QR code PNGs).
`templates/`	**Templates Subfolder:** Stores server-side rendering views (e.g., Thymeleaf HTML pages) for dynamic customer queue status.
`application.properties`	**Configuration File:** Global property file controlling server ports (e.g., `server.port=8080`) and environment parameters.

6. Testing Directory (`src/test/`)

File / Folder	Role & Responsibility
`src/test/java/com/queuenow/`	Test source root container for automated unit and component testing.
`QueuenowApplicationTests.java`	Context verification test ensuring Spring Boot initializes without configuration faults.